

Verification and Optimisation of Capacitive MEMS Displacement Sensor Simulation

Based on IEEE WINTeCHCON 2025 Reference Paper

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Abstract—This paper presents the verification and optimisation of a MEMS capacitive displacement sensor based on the reference IEEE WINTeCHCON 2025 paper titled “Design of Capacitive based MEMS Displacement Sensor using COMSOL”. The sensor employs a monolithic capacitive sensing probe fabricated using the Borosilicate Glass Reflow Process and incorporates a Kelvin Guard Ring to suppress fringe electric field effects and improve electric field uniformity.

The sensor geometry was resimulated in COMSOL Multiphysics 6.1 using electrostatic analysis and parametric sweep studies. Electric field norm distributions, electric potential distributions, and capacitance-versus-displacement characteristics were reproduced and analysed.

The resimulation demonstrated close agreement with the reference paper at 1 V excitation. However, discrepancies were identified in the 5 V capacitance results reported in the reference paper, which were found to contradict electrostatic theory.

A parametric optimisation study was performed by varying the Kelvin Guard Ring width and inner insulator width. The optimised structure, consisting of a widened Kelvin Guard Ring and reduced insulator spacing, reduced the deviation from ideal capacitance from 8.79% to 1.92%.

Index Terms—MEMS, capacitive displacement sensor, Kelvin guard ring, COMSOL Multiphysics, electrostatics, borosilicate glass reflow process, fringe field suppression

I. INTRODUCTION

Microelectromechanical Systems (MEMS) displacement sensors are widely used in industrial automation, aerospace systems, robotics, biomedical instrumentation, and precision metrology applications where accurate non-contact displacement measurement is required. Among the available sensing techniques, capacitive displacement sensing is preferred because of its high sensitivity, fast response time, low power consumption, and ability to detect extremely small displacements at micro- and nano-scale resolutions.

The operating principle of a capacitive displacement sensor is based on the variation of capacitance between conductive electrodes caused by a change in displacement or air gap. In an

ideal parallel-plate capacitor, the capacitance is inversely proportional to the separation distance between the plates. However, practical MEMS capacitive sensors experience fringe electric field effects near the electrode edges, resulting in non-uniform electric field distribution and deviations from ideal capacitance behaviour.

To minimise these edge effects, Kelvin Guard Ring structures are commonly employed. A Kelvin Guard Ring is a concentric electrode maintained at the same electric potential as the sensing electrode. This arrangement confines the electric field lines within the sensing region, improves electric field uniformity, reduces parasitic capacitance, and enables the sensing electrode to behave more closely to an ideal parallel-plate capacitor.

Recent advancements in MEMS fabrication technology have enabled compact and highly reliable capacitive displacement sensors using Glass-in-Silicon or Borosilicate Glass Reflow fabrication processes. In this technique, conductive silicon structures are embedded within insulating borosilicate glass, resulting in strong mechanical stability, excellent electrical isolation, and improved thermal compatibility.

The reference paper titled “Design of Capacitive based MEMS Displacement Sensor using COMSOL” presented the design and simulation of a MEMS capacitive displacement sensor using COMSOL Multiphysics. The sensor geometry, electric field distribution, and capacitance-displacement characteristics were analysed using electrostatic simulations.

In this work, the sensor presented in the reference paper was resimulated and analytically verified using COMSOL Multiphysics 6.1.

The objectives of this work are:

- 1) To reproduce the electric field norm and electric potential distributions reported in the reference paper.
- 2) To verify the capacitance-versus-displacement characteristics through electrostatic simulations.

- 3) To identify discrepancies between the reference paper and electrostatic theory.
- 4) To optimise Kelvin Guard Ring and insulator dimensions.
- 5) To propose an improved MEMS capacitive displacement sensor structure.

II. DEVICE STRUCTURE AND OPERATING PRINCIPLE

A. Materials and Geometry

The MEMS capacitive displacement sensor consists of a concentric three-electrode configuration fabricated using the Borosilicate Glass Reflow Process.

The materials used are:

- **Silicon** – used for the sensing electrode, guard ring, and ground ring.
- **Borosilicate Glass** – used as the insulating material.
- **Air** – acts as the dielectric medium between the sensing probe and target electrode.

The sensor geometry includes:

- Measuring electrode diameter (D)
- Inner insulator width (t_1)
- Kelvin Guard Ring width (w_1)
- Outer insulator width (t_2)
- Ground Ring width (w_2)
- Probe thickness (h)

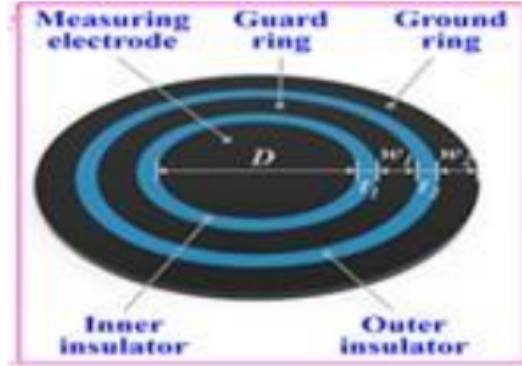


Fig. 1. Monolithic capacitive sensing probe showing sensing electrode, guard ring, and ground ring geometry.

B. Sensing Mechanism

The sensor operates as a variable-gap parallel-plate capacitor.

The capacitance is given by:

$$C = \epsilon_0 \frac{A}{d} \quad (1)$$

where:

- ϵ_0 = permittivity of free space
- A = effective electrode area
- d = separation distance between electrodes

As the displacement changes, the air gap changes, resulting in variation of capacitance.

C. Kelvin Guard Ring Principle

The Kelvin Guard Ring is maintained at the same potential as the sensing electrode. This arrangement forces the electric field lines to remain parallel within the sensing region, thereby minimising fringe-field effects.

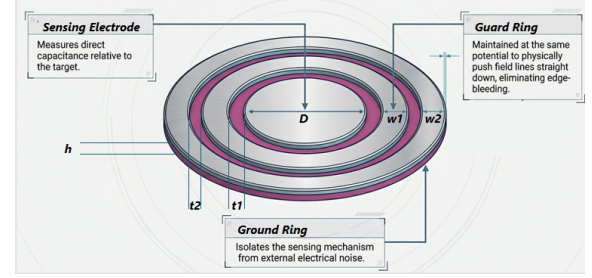


Fig. 2. Electrode configuration showing sensing electrode, guard ring, and displacement measurement mechanism.

III. FABRICATION PROCESS

A. MEMS Fabrication Steps

The MEMS capacitive displacement sensor is fabricated using the Borosilicate Glass Reflow Process on a silicon wafer substrate.

In the fabrication figure:

- Grey region represents silicon substrate.
- Pink region represents photoresist layer.
- White trench regions represent etched cavities.
- Gold/light yellow regions represent borosilicate glass.
- Dark grey/black pads represent metallisation layers.

The fabrication process proceeds as follows:

- 1) **Preparation:** Silicon wafer is cleaned and oxidised.
- 2) **Lithography:** Photoresist is applied and patterned.
- 3) **Etching:** Deep reactive ion etching (DRIE) creates silicon trenches.
- 4) **Stripping:** Photoresist and oxide layer removed via Buffered Oxide Etch (BOE).
- 5) **Glass Flow:** Borosilicate glass positioned over the etched area.
- 6) **Reflow & Grind:** Glass is reflowed into trenches; excess removed via top/bottom back grinding.
- 7) **Metallisation:** Metal deposited for sensor and ground electrode contact pads.

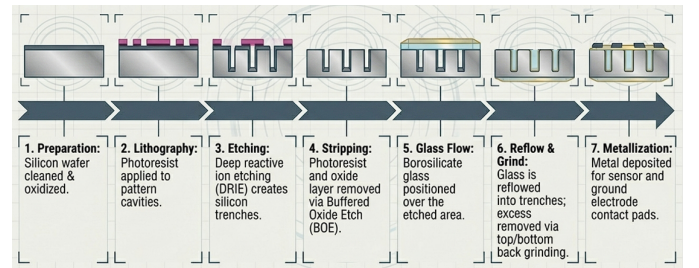


Fig. 3. Seven-step MEMS fabrication process using Borosilicate Glass Reflow technique.

B. Glass Reflow Process

Borosilicate glass is heated above its softening temperature, allowing it to flow into etched silicon trenches. Upon cooling, strong hermetic sealing and electrical isolation are achieved.

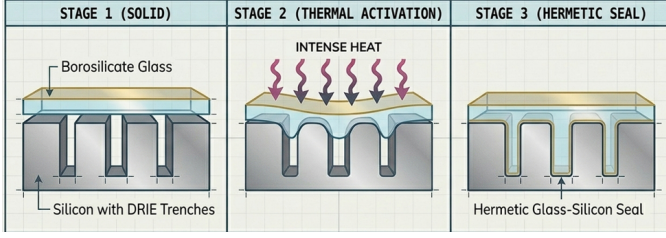


Fig. 4. Borosilicate Glass Reflow Process stages.

IV. SIMULATION ENVIRONMENT

A. COMSOL Setup

The sensor was simulated using COMSOL Multiphysics 6.1 with the Electrostatics module.

Simulation conditions included:

- 3-D sensor geometry
- Stationary electrostatic solver
- Parametric sweep from $10\ \mu\text{m}$ to $600\ \mu\text{m}$
- 1 V and 5 V applied to sensing electrode and guard ring
- Ground boundary applied to target electrode
- Fine mesh near electrode boundaries

B. Simulation Layout

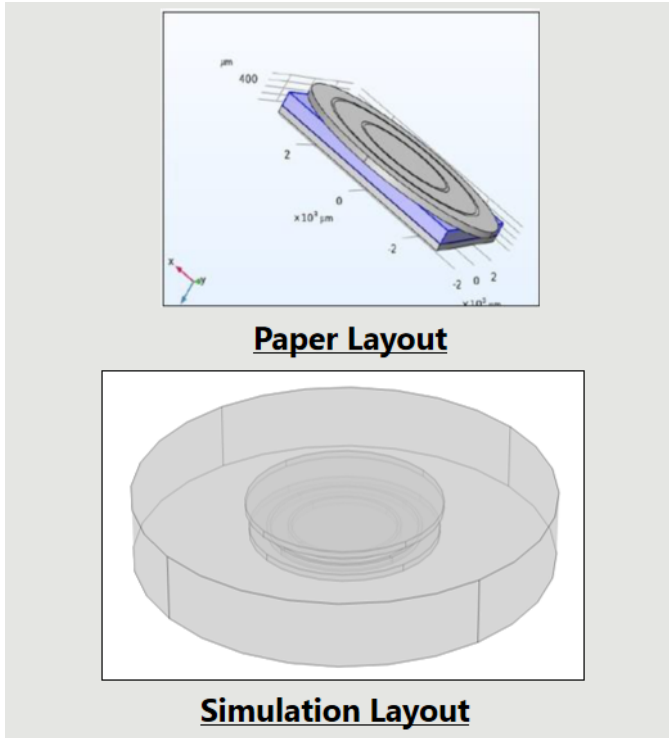


Fig. 5. COMSOL simulation layout and reference paper comparison.

V. SIMULATION RESULTS

A. Electric Potential Distribution

1) *Reference Paper Results:* The reference paper reported a concentric electric potential distribution with uniform potential near the sensing electrode.

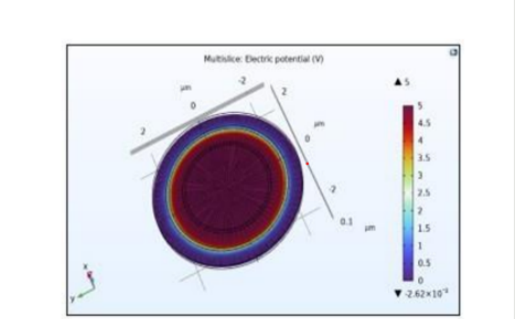


Fig. 4. Electric Potential Distribution

Fig. 6. Electric potential distribution from reference paper.

2) *Resimulation Results:* The resimulation successfully reproduced the electric potential distribution and equipotential behaviour.

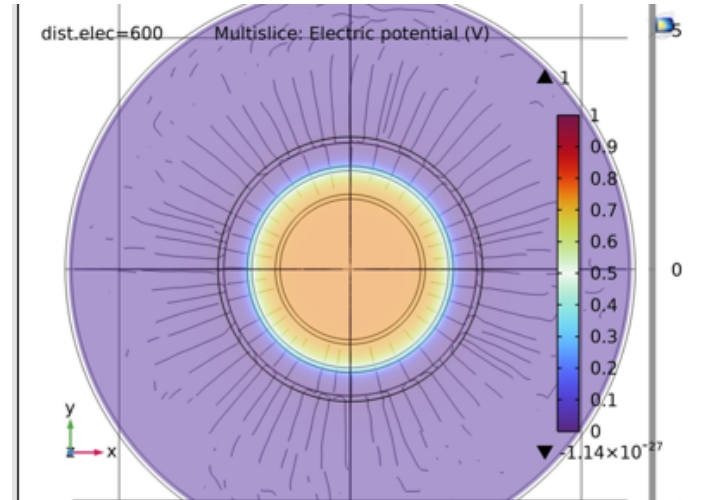


Fig. 7. Resimulated electric potential distribution.

B. Electric Field Norm

1) *Reference Paper Results:* The reference paper showed strong electric field concentration near electrode boundaries.

2) *Resimulation Results:* The resimulation reproduced the electric field distribution and demonstrated effective fringe-field suppression.

C. Capacitance vs Displacement at 1 V

The capacitance-versus-displacement characteristics closely followed the theoretical inverse relationship.

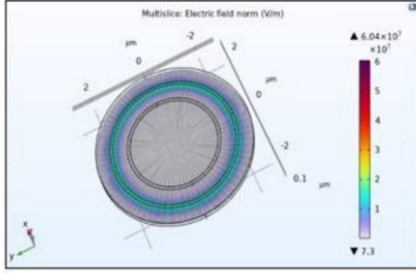


Fig. 5. Electric Field Norm Distribution

Fig. 8. Electric field norm distribution from reference paper.

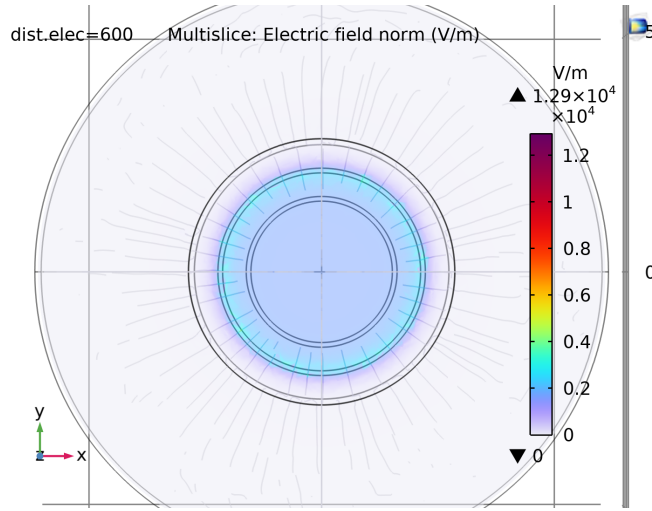


Fig. 9. Resimulated electric field norm distribution.

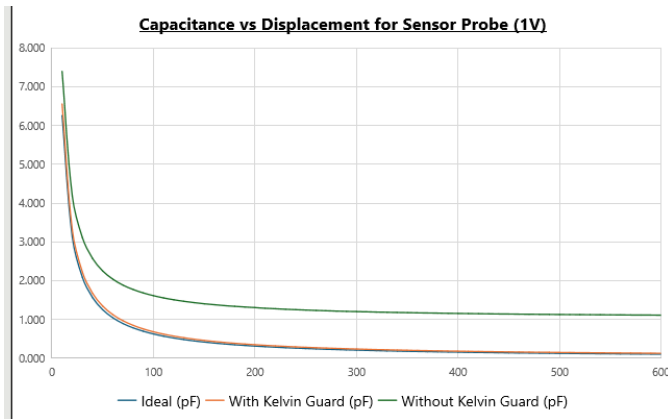


Fig. 10. Capacitance versus displacement at 1 V.

D. Capacitance vs Displacement at 5 V

The reference paper reported inconsistent capacitance values at 5 V excitation.

The resimulation verified that capacitance remains independent of voltage in linear dielectric media.

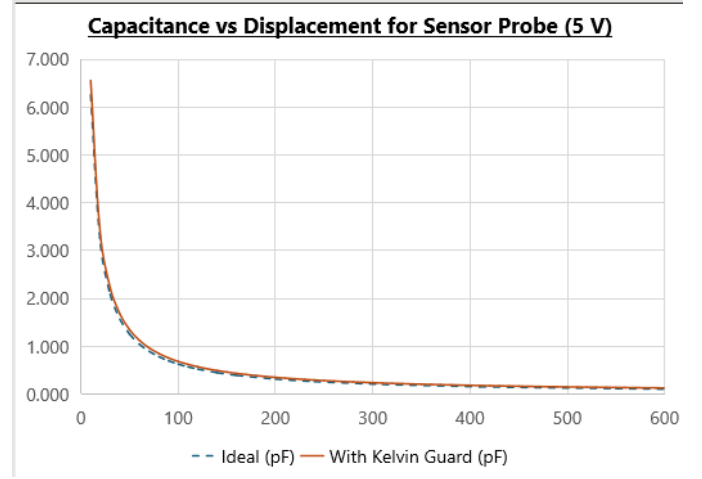


Fig. 11. Capacitance versus displacement at 5 V.

VI. PARAMETRIC STUDY AND GEOMETRY OPTIMISATION

A. Effect of Kelvin Guard Ring Width

Increasing the Kelvin Guard Ring width improved electric field confinement and reduced fringe-field deviation.

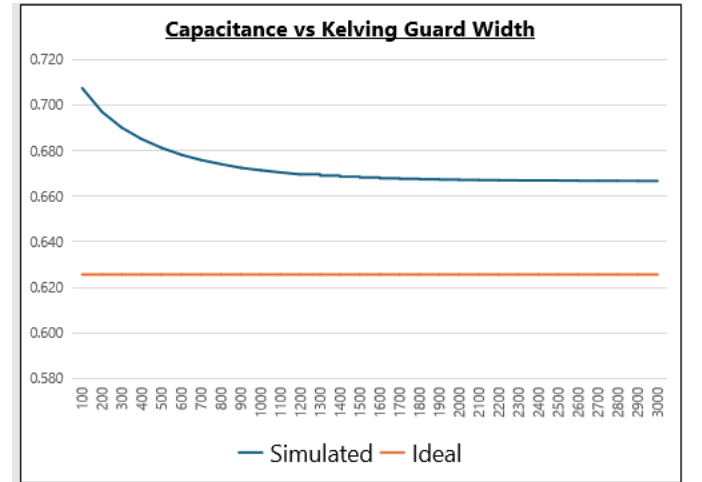


Fig. 12. Effect of Kelvin Guard Ring width on capacitance deviation.

B. Effect of Inner Insulator Width

Reducing inner insulator width moved the guard ring closer to the sensing electrode and improved capacitance accuracy.

VII. IMPROVED STRUCTURE

The improved structure was developed by:

- Increasing Kelvin Guard Ring width to 3000 μm
- Reducing inner insulator width to 30 μm

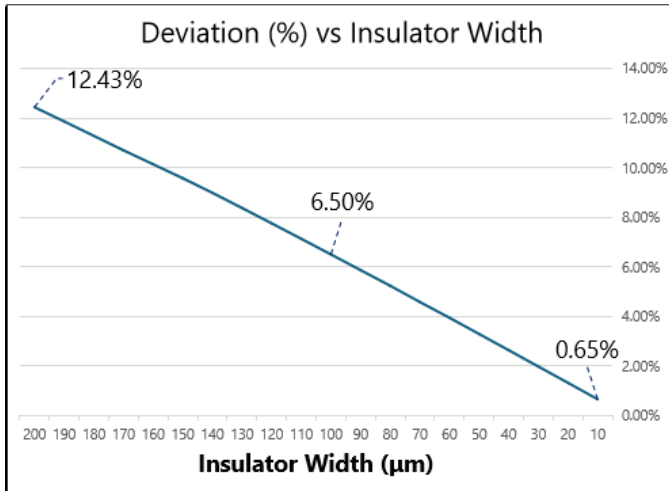


Fig. 13. Effect of inner insulator width on capacitance deviation.

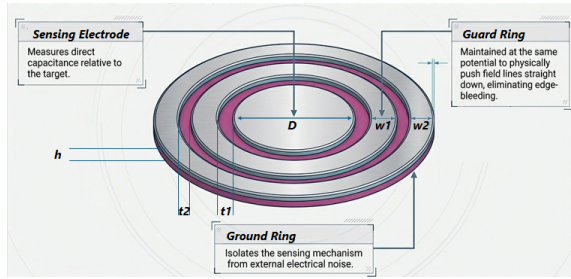


Fig. 14. Improved MEMS capacitive displacement sensor structure.

A. Improved Capacitance Results

The improved geometry significantly reduced capacitance deviation from ideal values.

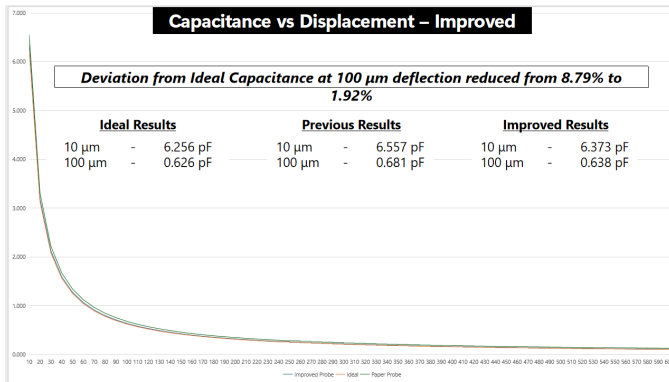


Fig. 15. Improved capacitance-versus-displacement comparison.

VIII. DISCUSSION

The resimulation performed in COMSOL Multiphysics successfully reproduced the major electrostatic characteristics reported in the reference paper, including the electric field norm distribution, electric potential distribution, and capacitance-displacement behaviour.

The electric field norm plots demonstrated that the Kelvin Guard Ring effectively confines the electric field within the sensing region and minimises fringe electric field effects near the sensing electrode boundaries.

The capacitance-versus-displacement results at 1 V closely matched both theoretical predictions and the reference paper results.

However, a significant discrepancy was identified in the reference paper's 5 V capacitance results. In linear dielectric media, capacitance should remain independent of applied voltage. The resimulation confirmed this expected behaviour, indicating that the discrepancy in the reference paper likely resulted from modelling inconsistencies or reporting errors.

The geometry optimisation study further demonstrated that wider Kelvin Guard Rings and reduced insulator widths significantly reduce fringe-field effects and improve capacitance accuracy.

IX. CONCLUSION

This work presented the verification and optimisation of a MEMS capacitive displacement sensor based on the IEEE WINTeCHCON 2025 reference paper.

The electric field norm distribution, electric potential distribution, and capacitance-displacement characteristics were successfully reproduced using COMSOL Multiphysics 6.1.

The Kelvin Guard Ring principle was verified to effectively suppress fringe electric field effects and improve sensing accuracy.

A discrepancy in the reference paper's 5 V capacitance results was identified and analytically explained.

Finally, geometric optimisation significantly improved sensor performance, reducing capacitance deviation from 8.79% to 1.92%.

ACKNOWLEDGMENT

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REFERENCES

- [1] K. N., C. K. P., and Y. G., "Design of Capacitive based MEMS Displacement Sensor using COMSOL," in *Proc. 2025 IEEE 6th International Women in Technology Conference (WINTeCHCON)*, 2025, pp. 1–6.
- [2] W. Zhang, Z. Bu, J. Li, Y. Wu, and L. Wang, "A New Fabrication Method for Capacitive Displacement Sensor," *IEEE Sensors Journal*, vol. 24, no. 9, pp. 13856–13865, May 2024.
- [3] J. Zhou, W. Zhang, Y. Li, X. Wang, L. Zhang, and H. Li, "Phase-Based Displacement Sensor With Improved Spatial Frequency Estimation and Data Fusion Strategy," *IEEE Sensors Journal*, vol. 22, no. 4, pp. 3306–3317, Feb. 2022.
- [4] C. Chen, L. Zhao, and H. Wang, "Design of a Sensor Structure for Linear Displacement Measurement Based on a Capacitive Grid Sensor," *IEEE Sensors Journal*, vol. 24, no. 19, pp. 30184–30194, Oct. 2024.
- [5] X. Liu, K. Peng, Z. Chen, H. Pu, and Z. Yu, "A New Capacitive Displacement Sensor With Nanometer Accuracy and Long Range," *IEEE Sensors Journal*, vol. 16, no. 8, pp. 2306–2315, Apr. 2016.
- [6] J. Lan, L. Gui, L. Lang, H. Ma, S. Zhu, and Q. Huang, "High-Precision Capacitive Displacement Sensor for Gravitational Wave Detection: Sensitive Probe Modeling, Analysis, and Optimization," *IEEE Sensors Journal*, vol. 24, no. 8, pp. 12514–12525, Apr. 2024.

- [7] K. Peng, Z. Deng, X. Liu, H. Wang, and Z. Yu, "Planar Two-Dimensional Capacitive Displacement Sensor Based on Time Grating," *IEEE Transactions on Industrial Electronics*, vol. 71, no. 4, pp. 4262–4273, Apr. 2024.
- [8] S. D. Senturia, *Microsystem Design*. Kluwer Academic Publishers, 2001.
- [9] COMSOL AB, *COMSOL Multiphysics Reference Manual*, Version 6.1, 2023.